

HOMELAB INFRASTRUCTURE PROJECT

Network, Identity, Storage & Data Protection Documentation

Domain: homelab.local (NetBIOS: HOMELAB)

Document Version 1.0 · Prepared September 2026

Consolidated and edited from project working notes

Table of Contents

- 1. Introduction..... 3
- 2. Project Scope 4
- 3. Network Architecture..... 6
- 4. Identity & Access Management (Active Directory) 9
- 5. Storage Architecture 10
- 6. Data Protection Strategy 13
- 7. Current Status & Outstanding Work..... 15
- Appendix A: Storage & Virtualization Glossary 16**
- Appendix B: Document Sources 17**

1. Introduction

This document consolidates the working notes produced over several build sessions of a self-hosted home laboratory (“homelab”) into a single, structured reference. The lab was built to gain hands-on, enterprise-style experience across virtualization, network security, identity management, file/block storage, and data protection — the same disciplines found in a small-to-mid-size business IT environment.

The environment is virtualized on a single VMware ESXi host and is organized around a Windows Server 2022 Active Directory domain (homelab.local), a pfSense firewall/router/VPN appliance, and a TrueNAS SCALE storage server providing both file-level (SMB/NAS) and block-level (iSCSI/SAN) storage to the lab.

The project was carried out iteratively — configuration, testing, and troubleshooting. This document reorganizes several separate sets of session notes (network setup, Active Directory and NAS permissions, SAN/iSCSI provisioning, capacity expansion, and data protection) into one coherent record, with a defined project scope, consolidated architecture diagrams, and end-to-end data flow diagrams added for clarity.

Purpose of this document

- Provide a single source of truth describing what has been built, how it is configured, and why specific design decisions were made.
- Present the architecture and data flows in a form suitable for review, onboarding, or portfolio use.
- Track outstanding work and known gaps so the project can be picked up cleanly in future sessions.

2. Project Scope

2.1 Objective

Build and document a small but realistic enterprise-style IT environment at home, covering network segmentation and perimeter security, centralized identity, file and block storage with access control, and a data-protection strategy — while capturing the reasoning and troubleshooting behind each decision.

2.2 In Scope

- Segmented network design: home network, an isolated Lab-LAN behind a virtual firewall, and a VPN tunnel range for remote access.
- pfSense deployment: initial WAN/LAN configuration, OpenVPN remote-access server, and firewall rules enabling controlled home-to-lab access.
- Active Directory domain (homelab.local): organizational units, domain users, security groups, and domain-joined clients.
- TrueNAS file-level storage (NAS/SMB): department-based datasets with group-based ACLs matching AD security groups.
- TrueNAS block-level storage (SAN/iSCSI): iSCSI target/extent/portal configuration and a VMFS datastore for ESXi.
- Live capacity expansion of a production-style storage pool, zvol, LUN, and datastore without data loss.
- Data protection: manual and scheduled ZFS snapshots, two verified recovery methods, and a replication strategy for off-pool resilience.

2.3 Out of Scope / Not Yet Implemented

- Fully working remote access from an external network (currently limited to the local office LAN — see Section 3.5).
- Joining the third Windows 10 client to the domain.
- Domain-wide logon-rights Group Policy Object (“Allow-All-Users-Logon”).
- A physically separate replication target (third TrueNAS SCALE box) — planned, not yet built.
- NFS shares (present on TrueNAS but intentionally left disabled).
- Formal performance benchmarking (IOPS / throughput / latency) of the iSCSI datastore.

2.4 Project Phases

Phase	Focus	Status
Phase 0	Network topology, pfSense firewall & OpenVPN remote access	Core config complete; remote (off-LAN) access pending

Phase	Focus	Status
Phase 1	Active Directory + TrueNAS SMB/NAS permissions	Complete
Phase 2	SAN / iSCSI storage + live capacity expansion	Complete
Phase 3	Data protection: snapshots, recovery, replication planning	Snapshots complete; replication target build pending

3. Network Architecture

The lab uses three distinct address ranges to separate the home network, the isolated lab environment, and remote VPN clients. All lab infrastructure runs as virtual machines on a single ESXi host that itself sits on the home network.

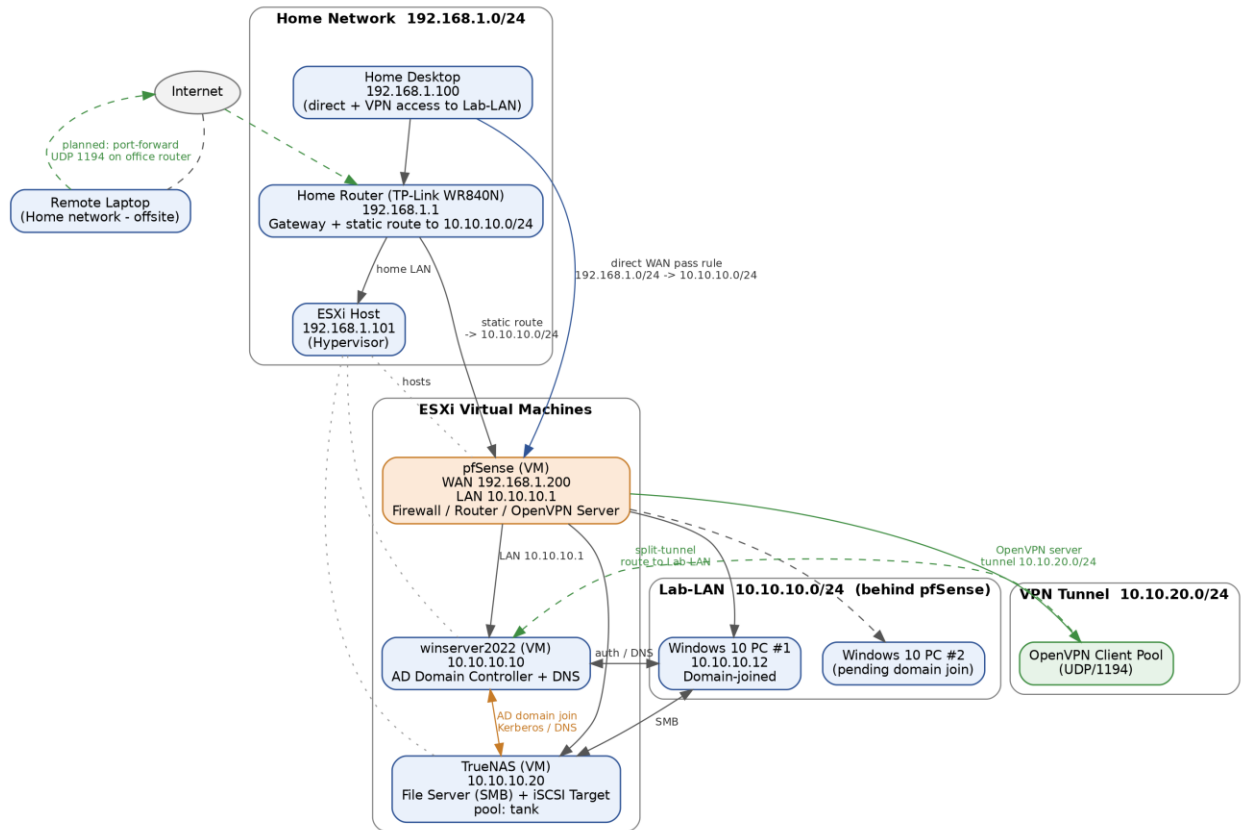


Figure 1. Overall network topology — home network, ESXi-hosted VMs, isolated Lab-LAN, and VPN tunnel.

3.1 Network Segments

Segment	Subnet	Purpose
Home network	192.168.1.0/24	Home router, home desktop, physical laptop, ESXi host
Lab-LAN	10.10.10.0/24	Isolated ESXi lab network, behind pfSense
VPN Tunnel	10.10.20.0/24	Address pool for remote OpenVPN clients

3.2 Key Hosts

Device	IP	Role
Home router (TP-Link WR840N)	192.168.1.1	Home network gateway
pfSense (VM)	WAN 192.168.1.200 / LAN 10.10.10.1	Firewall, router, VPN server between home and Lab-LAN

Device	IP	Role
winserver2022 (VM)	10.10.10.10	Active Directory Domain Controller + DNS
TrueNAS (VM)	10.10.10.20	File server (SMB) + iSCSI target, domain-joined
ESXi host	192.168.1.101	Hypervisor; sits directly on the home network
Windows 10 PC #1	10.10.10.12	Domain-joined test client
Windows 10 PC #2	(pending)	Domain-joined test client — not yet joined
Home desktop	192.168.1.100	Regular home PC; given direct + VPN access to Lab-LAN

3.3 pfSense Configuration

Initial setup

- WAN interface: static, on the home network (192.168.1.200/24, gateway 192.168.1.1).
- LAN interface: Lab-LAN (10.10.10.1/24).
- “Block RFC1918 Private Networks” and “Block bogon networks” were temporarily unchecked during initial local testing, then re-enabled for normal operation (required because a private-range source IP is otherwise blocked on WAN).
- The default pfSense admin password has not yet been changed — the setup warning banner still persists (see Section 7).

OpenVPN remote-access server

- CA: HomeLab-CA (RSA 2048-bit, 3650-day lifetime); Server cert: HomeLab-Server-cert (2048-bit, 398-day lifetime, AES-256-GCM).
- Interface: WAN · Protocol: UDP/IPv4 · Port: 1194 (default).
- Tunnel network: 10.10.20.0/24 · Local network pushed to clients: 10.10.10.0/24 (routes VPN clients into the Lab-LAN).
- Redirect Gateway unchecked — split tunneling, so only lab-bound traffic travels over the VPN.
- DNS Server 1 pushed to clients: 10.10.10.1. Firewall rules were auto-created by the wizard (WAN pass for UDP/1194, OpenVPN interface pass-through).
- VPN user: vpnadmin, a local pfSense account (admins group) with a client certificate signed by HomeLab-CA. The OpenVPN Client Export package required upgrading pfSense 2.6.0 → 2.7.0 first (its repo needed a newer PHP major version).

3.4 Remote Access — Status and Data Flow

An important operational detail: the lab is physically located in an office, while the intended “remote” client (a laptop) stays at home on an entirely different network. Local (same-LAN) VPN testing has succeeded, but true remote access from an external network has not yet been established.

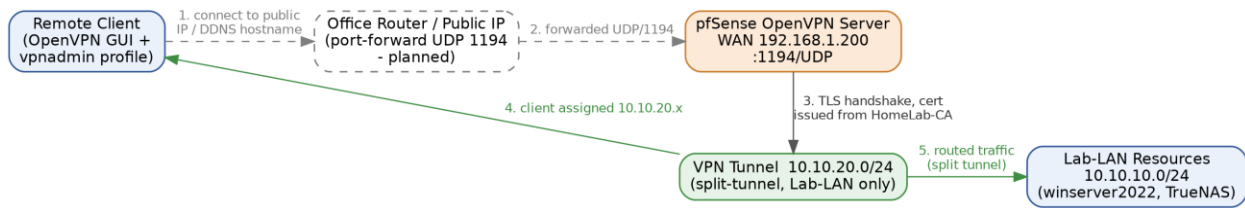


Figure 2. Remote-access (OpenVPN) data flow. Steps 1–2 (dashed) are not yet operational off-site.

- Local test result: client connected, was assigned 10.10.20.2, and successfully pinged both 10.10.10.1 (pfSense) and 10.10.10.10 (winserver2022) through the tunnel.
- Gap: the OpenVPN client config currently points at 192.168.1.200, which is only reachable from inside the office LAN itself.
- To enable genuine remote access: forward UDP/1194 on the office router to pfSense’s WAN IP, then re-export the client config to use the office’s public IP or a DDNS hostname instead of the private WAN address.
- After local testing, “Block private networks” and “Block bogon networks” were correctly re-enabled on WAN — this does not affect a genuine remote client connecting from a public IP.

3.5 Direct Home-to-Lab Access (non-VPN)

A thirdary goal was letting the home desktop (192.168.1.x) reach the Lab-LAN (10.10.10.x) directly, without running a VPN client.

- A WAN firewall pass rule was added (Source 192.168.1.0/24 → Destination 10.10.10.0/24), positioned above the block rules — pfSense evaluates rules top-down, first match wins.
- Root cause found: even with the firewall rule correct, traffic never reached pfSense because the home router had no route telling it how to reach 10.10.10.0/24.
- Fix: a static route was added on the TP-Link home router (Destination 10.10.10.0/255.255.255.0, Gateway 192.168.1.200, enabled on LAN & WLAN).
- End-to-end testing from the actual physical home desktop was still in progress as of the last update; earlier ping attempts were mistakenly run from the ESXi console / a lab VM, which does not exercise the WAN-rule path.

4. Identity & Access Management (Active Directory)

winserver2022 (10.10.10.10) is the domain controller and authoritative DNS server for homelab.local. It self-references its own DNS (127.0.0.1) and forwards external lookups to 10.10.10.1 (pfSense) and 8.8.8.8, since a DC's own DNS has no forwarders configured by default and otherwise cannot resolve internet names.

4.1 Organizational Units

OUs organize AD objects only — they do not themselves carry permissions:

- IT
- Finance
- Backup

4.2 Domain Users

OU	Users
IT	james (james@homelab.local), Alex, Shamie — 3 users total
Finance	purchases (purchases@homelab.local), plus 2 more — 3 users total

Note: james and purchases also exist as separate local Windows accounts on other machines. These are unrelated identities — local accounts do not carry over into AD.

4.3 Security Groups

Security groups (not OUs) carry the actual permissions referenced by TrueNAS ACLs:

Group	OU Location	Members
it-access	IT	3 IT domain users + Domain Admins
finance-access	Finance	3 Finance domain users + Domain Admins
backup-access	Backup	James, Shamie + Domain Admins

Domain Admins was added as a member of all three access groups, so any Domain Admin inherits access via group nesting.

4.4 Domain-Joined Clients & Logon Model

- Windows 10 PC #1 (DESKTOP-6S3I8N7): joined to homelab.local. Its DNS server had to be corrected from 127.0.0.1 / 8.8.8.8 to 10.10.10.10 — client machines must point to the DC for DNS; only the DC itself uses 127.0.0.1. HOMELABJames was added as a Standard account during the join wizard.
- Windows 10 PC #2: not yet joined.
- By default, only the domain user added during the join wizard (and Domain Admins) can log into a freshly-joined PC. A domain-wide GPO (“Allow-All-Users-Logon”) granting Domain Users

the “Allow log on locally” right is planned so any of the 6 domain users can log into either PC without per-machine configuration — see Section 7.

5. Storage Architecture

TrueNAS (10.10.10.20, pool tank) provides both file-level storage (NAS, over SMB) and block-level storage (SAN, over iSCSI) to the lab, from the same underlying ZFS pool.

5.1 NAS / File-Level Storage (SMB)

Datasets were restructured from a single generic dataset into a department-based layout, each created with the SMB dataset preset (Case Sensitivity: OFF):

Dataset	Path	SMB Share
tank/IT	/mnt/tank/IT	IT
tank/Finance	/mnt/tank/Finance	Finance
tank/Backup	/mnt/tank/Backup	Backup

Final ACL structure (applied identically to all three datasets)

- owner@ – root – Allow | Full Control
- Group – matching access group (it-access / finance-access / backup-access) – Allow | Full Control
- Group – builtin_administrators – Allow | Full Control
- Group – HOMELAB\domain admins – Allow | Full Control (added explicitly on the NAS side, in addition to AD-side group nesting)
- ACL flags left at Inherit, so new files/folders automatically pick up the same permissions.

Key fix applied: the default Group – builtin_users entry (Allow | Modify) was removed from all three datasets. Leaving it in place would have let any authenticated domain user access any share, defeating the purpose of group-based restriction.

SMB service

- Status: Running, starts automatically · Workgroup: HOMELAB · NetBIOS name: truenas
- SMB1 support and NTLMv1 auth left disabled (secure defaults)

Client-side verification

- Mapped drive Z: on winserver2022 → \\10.10.10.20\IT, connected as james → success (confirms it-access membership grants access via the ACL).
- Still pending: formal negative-access test confirming james is denied access to \\10.10.10.20\Finance.
- Note: browsing via Network → TRUENAS → share in File Explorer uses whichever account is currently logged into Windows (no credential prompt) — different from “Map network drive → Connect using different credentials,” which explicitly authenticates as the specified user.

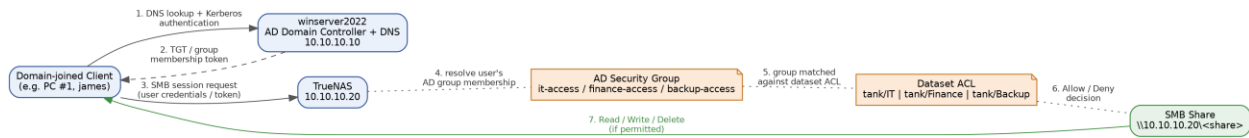


Figure 3. Authentication & file-access data flow for an SMB share request.

5.2 SAN / Block-Level Storage (iSCSI)

Goal: connect ESXi to TrueNAS block storage over iSCSI and create a VMFS datastore for VMs.

Naming convention: homelab-iscsi1, homelab-iscsi2, etc. for successive extents/zvols.

TrueNAS-side configuration

- Zvol: tank/homelab-iscsi1 — initial size 20 GiB, Thick provisioning, unencrypted, path /dev/zvol/tank/homelab-iscsi1.
- iSCSI Extent: homelab-iscsi1, Device-backed by the zvol above; Sharing Platform preset “VMware: Extent block size 512b, TPC enabled, no Xen compat mode, SSD speed.”
- iSCSI Target: homelab-iscsi1 (created via Wizard). Portal: 10.10.10.20, port 3260. Initiators left unrestricted (acceptable for a homelab).
- iSCSI Service: Enabled, starts automatically, status Running.

Troubleshooting note: the first iSCSI Wizard attempt failed with [EINVAL] – the target zvol device did not exist, because the wizard’s inline “Create New” zvol flow silently failed to create it (a Pool/Dataset selection was reset mid-flow). Fix: the zvol was created manually via Datasets → tank → Add Zvol, then the wizard was re-run and the existing zvol selected from the Device dropdown.

ESXi-side configuration

- Software iSCSI adapter enabled: vmhba64. Initiator IQN: iqn.1998-01.com.vmware:localhost:1764664256:64 (iscsi_vmk).
- CHAP authentication: not used.
- Dynamic target 10.10.10.20:3260 added; after Save + Rescan, ESXi discovered the LUN as “TrueNAS iSCSI Disk” (naa.6589cfc000000dfa02b1f2345669252f), reporting “Normal, Degraded” — expected for a single-NIC homelab with no multipathing, not a fault.

Datastore creation

- New Datastore wizard run against the discovered LUN. Name: homelab-datastore1 (later renamed homelab-iscsi1 datastore). Partitioning: full disk. Filesystem: VMFS 6.
- Result: 19.75 GB usable (~20 GB minus VMFS overhead). Thin provisioning is supported at the datastore layer, even though the underlying zvol is Thick — so no real space savings occur unless the zvol itself is made Sparse.

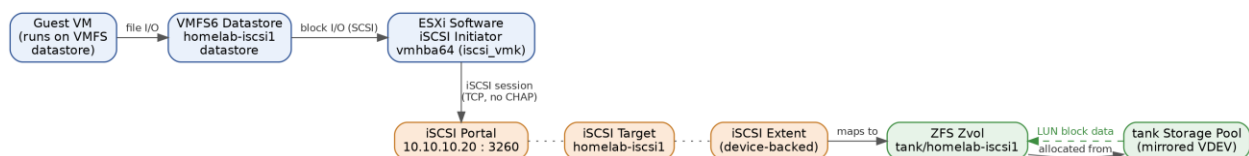


Figure 4. Block-storage (iSCSI) data flow, from a guest VM down to the physical storage pool.

5.3 Capacity Expansion Case Study (20 GB → 100 GB)

A live, in-place expansion of the iSCSI storage chain was carried out end-to-end without data loss, illustrating that every layer between the physical disk and the VM datastore must be grown explicitly.

Step	Action	Outcome
1	Grow both mirrored virtual disks on the TrueNAS VM (100 GB → 200 GB each) in ESXi; restart the VM	Both disks report 200 GB at the device level
2	First pool Expand attempt in TrueNAS	Failed — sgdisk could not grow the GPT partition (safe failure, no changes made)
3	Diagnose via TrueNAS Shell	GPT backup header still anchored at the old 100 GB boundary; only ~1 GiB free reported
4	Relocate GPT backup header on each disk (sgdisk -e, then partprobe)	Both disks confirm ~101 GiB free space, no reboot required
5	Retry pool Expand	Success — usable capacity rose from 95.44 GiB to 191.34 GiB; scrub completed in 7s, 0 errors
6	Resize zvol tank/homelab-iscsi1	20 GiB → 100 GiB, completed without warning (now well under the 80% pool-usage threshold)
7	Rescan + Increase Capacity on the ESXi iSCSI device and datastore	Datastore expanded to the full 100 GB, no leftover unallocated space

Key lessons learned

- Resizing a virtual disk in ESXi does not automatically grow anything above it — each layer (virtual disk → GPT partition → ZFS pool → zvol → iSCSI LUN → VMFS datastore) needs its own explicit “grow” action.
- A failed “Expand” is not necessarily destructive — TrueNAS refused to save changes when sgdisk errored, leaving the pool exactly as it was.
- The specific failure mode (GPT backup header stuck at the old disk-size boundary) is a known, safe, fixable situation via sgdisk -e followed by partprobe — no reboot required.
- Mirrored VDEVs require both members to be resized equally, since ZFS mirror capacity is bound by the smaller of the two disks.
- The iSCSI Wizard’s inline “Create New” zvol shortcut can silently fail; creating the zvol manually first and selecting it as an existing device is more reliable.

6. Data Protection Strategy

Data protection was implemented in three layers: one-off manual snapshots for milestone checkpoints, automated periodic snapshots for rolling recovery points, and a planned replication target for protection against loss of the entire storage pool.

6.1 Manual Snapshots

A manual snapshot was taken of each key storage object via Storage → Datasets → Snapshots → Add. Manual snapshots have no expiry — they persist until explicitly deleted.

Dataset / Zvol	Snapshot Name	Retention
tank/IT	IT-manual-2026-09-02_12-34	Not automatically destroyed
tank/Finance	Finance-manual-2026-09-02	Not automatically destroyed
tank/Backup	Backup-manual-2026-09-02	Not automatically destroyed
tank/homelab-iscsi1	homelab-iscsi1-manual-2026-09-02_12-46	Not automatically destroyed

6.2 Recovery Methods — Two Methods Verified

Method	What it does	Best used when
Copy from .zfs/snapshot/	Restores specific files only; nothing else in the dataset is touched	A small, known set of files was lost or changed
Rollback	Reverts the ENTIRE dataset to the snapshot's point in time; all later changes are discarded	Undoing everything since a known-good checkpoint

File-level restore was verified by deleting live files from \\10.10.10.20\IT, then copying them back from the hidden read-only path \\10.10.10.20\IT\.zfs\snapshot\IT-manual-2026-09-02_12-34\. Full rollback was separately verified on tank/IT using the safe default option “Stop Rollback if Newer Intermediate, Child, and Clone snapshots exist”, confirming the entire dataset state was restored to the snapshot's creation time.

6.3 Automated Protection — Periodic Snapshot Tasks

A Periodic Snapshot Task was configured for each dataset, taking a full daily snapshot at 00:00 with a defined retention window after which older snapshots are automatically deleted:

Dataset / Zvol	Naming Schema	Schedule	Keep For
tank/IT	IT-%Y-%m-%d_%H-%M	Daily @ 00:00	1 month
tank/Finance	Finance-%Y-%m-%d_%H-%M	Daily @ 00:00	1 month
tank/Backup	Backup-%Y-%m-%d_%H-%M	Daily @ 00:00	2 weeks

Dataset / Zvol	Naming Schema	Schedule	Keep For
tank/homelab-iscsi1	auto-%Y-%m-%d_%H-%M	Daily @ 00:00	1 month

- Each daily snapshot captures the full dataset state at that moment, not just files added that day.
- Retention is a rolling window — a snapshot is deleted exactly N days/weeks after it was created, so the pool always holds a running set of recent recovery points.
- If a file is deleted and not recovered before the oldest snapshot containing it expires, it becomes permanently unrecoverable via snapshots unless it also exists in a no-expiry manual snapshot.
- Manual and automated snapshots coexist independently in the same snapshot list, each with its own retention rule.

6.4 Replication — Concept and Planning

Replication protects against loss of the entire tank pool — something snapshots alone cannot do, since every snapshot still lives on the same pool as the original data. Replication sends an independent copy of snapshot data to a separate destination, fulfilling the “different media / a copy elsewhere” principle of the 3-2-1 backup rule.

Option Considered	Verdict
Replicate to a new dataset on the same tank pool	Rejected — same pool, same single point of failure
Replicate to the existing homelab-iscsi1 zvol/datastore	Rejected — same TrueNAS VM and same tank pool as the source
Deploy a third TrueNAS VM on the same ESXi host	Not currently possible — host has only ~1.5 GB RAM free; TrueNAS SCALE needs 8 GB+
Replicate to the Windows 10 home desktop via Rsync	Viable fallback — real independent hardware, but file-level only (Windows cannot receive ZFS send streams)
Build a third, dedicated bare-metal machine running TrueNAS SCALE	SELECTED — genuine hardware independence and native ZFS send/receive replication

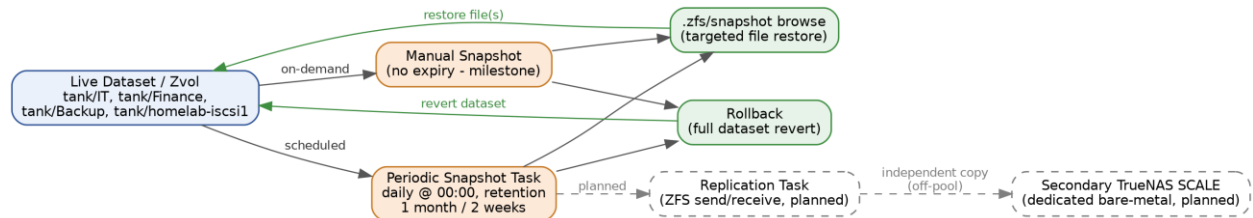


Figure 5. Data protection flow — snapshot creation, recovery paths, and planned replication.

7. Current Status & Outstanding Work

7.1 Network & Remote Access

- Change pfSense's default admin password (still on default — warning banner persists).
- Confirm home desktop → Lab-LAN direct access works end-to-end now that the static route is in place (test from the real physical home desktop, not a lab VM).
- Set up genuine remote access from home: port-forward UDP 1194 on the office router to pfSense's WAN IP, obtain a public IP or Dynamic DNS hostname, and re-export the OpenVPN client config to use it.

7.2 Identity & Access

- Join the third Windows 10 PC to homelab.local.
- Create the domain-wide GPO (“Allow-All-Users-Logon”) granting Domain Users the “Allow log on locally” right.

7.3 Storage

- Confirm the SMB shares list on TrueNAS shows all three shares correctly.
- Map network drives from a domain-joined client as test users; verify read/write/delete on the correct share.
- Negative permission test: confirm a user without access to a given share (e.g., james reaching \\10.10.10.20\Finance) is properly denied.

7.4 Data Protection

- Prepare a third desktop as bare-metal hardware (64-bit CPU, 8 GB+ RAM, dedicated disk for the ZFS pool).
- Install TrueNAS SCALE on the third machine and configure networking (static IP on 192.168.1.0/24), confirming reachability from 10.10.10.20.
- Create a receiving pool and destination dataset (e.g., backup-pool/IT-replica) and set up SSH trust between the two TrueNAS systems.
- Create Replication Tasks sending tank/IT, tank/Finance, and tank/Backup snapshots to the new box, then verify replicated data is browsable and restorable.

Appendix A: Storage & Virtualization Glossary

Terms below are mapped directly to how they were encountered in this project.

Term	How it applies in this project
SMB (Server Message Block)	The protocol behind the IT, Finance, and Backup shares on TrueNAS. Mapping \\10.10.10.20\IT as james used SMB.
NAS (Network-Attached Storage)	TrueNAS's file-sharing role — accessing \\10.10.10.20\IT through Windows Explorer uses TrueNAS as a NAS.
NFS (Network File System)	Not used in this project; seen listed alongside SMB on the TrueNAS Shares page (Stopped) as the Linux/Unix equivalent.
SAN (Storage Area Network)	The role TrueNAS took on for Phase 2 — handing ESXi a raw block device over the network instead of a folder.
iSCSI	The protocol used to make that SAN connection work: target homelab-iscsi1 at portal 10.10.10.20:3260, discovered by ESXi's software initiator vmhba64.
LUN (Logical Unit Number)	The identifier for the block of storage handed over — the zvol, discovered by ESXi as “TrueNAS iSCSI Disk.”
RAID	Not directly configured, but the tank pool's disks are arranged as a 2-wide Mirror — ZFS's version of RAID 1.
Storage Pool	The tank pool — the mirrored disks that all datasets (IT, Finance, Backup, homelab-iscsi1) are carved out of.
Datastore	The VMFS-formatted destination on ESXi, sitting on top of the iSCSI LUN, where VM files live.
VMFS	The filesystem ESXi used to format the LUN — VMFS 6 was selected during datastore creation.
Thin Provisioning	Supported at the VMFS layer, but the underlying zvol was created Thick, so the full size is already reserved on the pool regardless of VM-level usage.
IOPS / Throughput / Latency	Not yet measured in this project — relevant to future benchmarking of the iSCSI datastore.
Snapshot	A ZFS point-in-time capture, available for both SMB datasets and the iSCSI zvol.
Backup	The Backup OU, backup-access group, and Backup share/dataset are named for this purpose; a scheduled protection job now exists via Periodic Snapshot Tasks.
Replication	Sending snapshot data to a third, independent TrueNAS system — planned but not yet built.

Appendix B: Document Sources

This document consolidates and edits the following original working notes into a single reference:

- Home Lab Network Documentation — network topology, pfSense, OpenVPN, direct access.
- Homelab Configuration Notes — Active Directory + TrueNAS SMB permissions (Phase 1) and initial SAN/iSCSI setup (Phase 2).
- Homelab Data Storage Documentation — consolidated NAS/SMB and SAN/iSCSI documentation, plus the capacity expansion case study (Phase 3 sizing).
- Home Lab Documentation — Part 3 (Data Protection) — snapshots, recovery methods, and replication planning.
- Storage Vocabulary Mapped to Your Homelab — glossary of storage/virtualization terms as encountered in the project.

Architecture and data-flow diagrams throughout this document were newly created to visually consolidate information that was previously spread across the notes above.